Virtual Cell Models Inflate Perturbation Effect Sizes and Undermine Causal Gene Regulatory Network Recovery

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Abstract

Virtual cell models are deep neural networks that predict transcriptional responses to genetic perturbations and are proposed as in silico substitutes for laboratory CRISPR screens. Ahlmann-Eltze & Huber (2025) recently showed that current models fail to outperform simple linear baselines at per-gene prediction; we ask whether the same holds downstream when their predictions are used to recover causal gene regulatory networks. We introduce PerturbCausal, a benchmark that pushes predictions from four virtual cell models (GEARS, CPA, Geneformer, and the 2025 Arc Institute STATE foundation model) through four causal discovery algorithms (GIES, inspre, PC, NOTEARS) on the Replogle K562 and RPE1 Perturb-seq datasets and the Norman 2019 K562 dataset. Across five seeds, the three pre-2025 models do not significantly outperform a sparse linear regression trained on control cells alone. STATE is the first model in the roster to materially exceed the linear baseline ($F_1 = 0.462$), but its absolute causal F1 remains below 0.5, leaving more than half of real interventional edges unrecovered. We reconcile two competing narratives in the recent literature, mode collapse (Mejia et al., 2025) and effect-size inflation, as two views of the same failure: deep models predict similar dense response patterns across perturbations. A model-agnostic quantile-matching calibration corrects this magnitude failure exactly but recovers only about 40% of the causal gap, locating the remainder in joint-distribution structure that distributional calibration cannot reach. The benchmark, calibration toolkit, and reproducibility code are released as an open-source Python package.

1. Introduction

Virtual cell models (VCMs) are deep generative or predictive models of single-cell transcriptional response under perturbation (Lotfollahi et al., 2023; Roohani et al., 2024; Theodoris et al., 2023; Cui et al., 2024; Bunne et al., 2024; Arc Institute, 2025). If a VCM's predictions could be used in place of real CRISPR perturb-seq, the cost of mapping a gene regulatory network would collapse by orders of magnitude. The most consequential evaluation of a VCM is therefore the substitutability question: do downstream pipelines that consume real perturb-seq still produce the same outputs when fed predicted perturb-seq?

The recent finding of Ahlmann-Eltze & Huber (2025) answers this question no for the simplest downstream task — per-gene reconstruction — where five deep foundation models and two other neural predictors all fail to beat a sparse linear baseline. We extend their evaluation in two directions. First, we ask whether the substitutability gap persists when the downstream task is harder: structural recovery of a causal gene regulatory network from interventional data, the canonical use case for perturb-seq (Chevalley et al., 2025; Callahan et al., 2025). Second, we test whether a 2025-era foundation model with a substantially larger pretraining corpus and cell embedding (STATE (Arc Institute, 2025)) closes the gap.

Contributions. (i) The first benchmark of substitutability for causal recovery: four virtual cell models × four causal discovery algorithms (GIES (Hauser & Bühlmann, 2012), inspre (Brown et al., 2025), PC (Spirtes & Glymour, 2000), NOTEARS (Zheng et al., 2018)) on three Perturb-seq datasets. (ii) A reconciliation of two opposing narratives in the recent literature — mode collapse (Mejia et al., 2025) and effect-size inflation — as two views of the same failure mode. (iii) A quantile-matching calibration that closes the

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magnitude piece of the gap exactly but exposes a joint-distribution residue. (iv) The first integration of the STATE foundation model into a causal-recovery benchmark, with the full reproducibility pipeline (the released checkpoint requires three framework-versioning patches we document and resolve).

Related work. Three recent results bear directly on this paper. Ahlmann-Eltze & Huber (2025) benchmark five foundation models and two other deep predictors against linear baselines for per-gene perturbation prediction and find no deep model wins; their result targets distributional reconstruction, ours targets downstream structural recovery, and the two are not redundant (predictions can fail one task and pass the other, or vice versa). Mejia et al. (2025) identify a mode-collapse signature in single-cell perturbation models and propose diversity-promoting training losses; we show this signature and the seemingly-contradictory effect-size-inflation signature are equivalent at the population level. Chevalley et al. (2025) introduce CausalBench, a large-scale benchmark of causal discovery algorithms on Perturb-seq data; our work is complementary, scoring VCM predictions against the algorithms' real-data outputs rather than scoring algorithms against each other. The closest framing of our question appears in the NeurIPS 2025 perspective of Callahan et al. (2025), who outline an evaluation philosophy for virtual cells as causal world models but do not publish a benchmark.

2. Methods

Datasets. K562 and RPE1 Perturb-seq from Replogle et al. (2022) and K562 CRISPRa perturb-seq from Norman et al. (2019), downsampled to $N = 200$ perturbation-eligible genes (per-perturbation cell counts ≥ 10).

Virtual cell models. GEARS (Roohani et al., 2024) (graph-augmented MLP, hidden 64, 20 epochs), CPA (Lotfollahi et al., 2023) (NB-loss VAE, latent 64, encoder/decoder 3 layers, 300 epochs), Geneformer V2 (Theodoris et al., 2023) (104M-parameter transformer, fine-tuned in-silico perturbation), and STATE (Arc Institute, 2025) (SE-600M cell encoder + ST-SE-Replogle perturbation model, run zero-shot from the released zeroshot/k562 checkpoint).

ElasticNet baseline. A per-gene sparse linear regression ($\alpha = 0.1, l_1$-ratio $= 0.5$) trained on control cells only and applied by zeroing the perturbed gene's expression. This is the same family of baseline that Ahlmann-Eltze & Huber (2025) found to match deep methods on per-gene reconstruction.

Method	Causal F1
ElasticNet	0.426 ± 0.016
CPA	0.412 ± 0.016
GEARS	0.406 ± 0.015
Geneformer	0.425 ± 0.011
STATE	$0.462 \pm$—
GRNBoost2 (random reference)	0.098 ± 0.004

Table 1. Causal F1 against real-data GIES ground truth on Replogle K562 at N=200. Mean ± s.d. across five seeds (42, 123, 456, 789, 1024), except STATE (single-seed; 320-cell subset due to CPU-bound SE-600M cost). The random reference (GRNBoost2) confirms all methods recover non-trivial signal; pairwise differences among the four predictive methods other than STATE are not significant ($p > 0.05$, paired permutation). Full per-method precision, recall, and AUPRC are in the appendix.

Overlay sampling for distributional matching. For each perturbation, 100 synthetic cells are drawn by bootstrap-resampling control cells, multiplying by the predicted per-gene ratio (clipped to $[0.01, 10]$), and Poisson-resampling. This preserves per-cell control covariance while imposing the predicted shift in mean.

Causal discovery. GIES (Hauser & Bühlmann, 2012) runs on ∼20-gene partitions of the overlay-sampled expression matrix (gauss_int_l0_pen score). inspre (Brown et al., 2025) runs the full λ-path on the average causal effect (ACE) matrix. PC and NOTEARS run on per-perturbation ACE.

Calibration. Per-model quantile matching against the empirical real-data $|\log_2 \mathrm{FC}|$ distribution, applied post-hoc to predicted means before overlay sampling.

Evaluation. F1, precision, recall against real-data GIES edges (the same algorithm on the real Perturb-seq), reported at seed = 42 and across $n = 5$ seeds (42, 123, 456, 789, 1024) for CIs.

3. Results

The substitutability gap is real but real-data GIES is the ceiling. On Replogle K562 at N=200, ElasticNet, CPA, GEARS, and Geneformer cluster in the F1 range 0.41–0.43 with overlapping intervals (Table 1). A 104M-parameter pretrained transformer (Geneformer) is statistically indistinguishable from a sparse linear regression with $\sim 200^2$ coefficients. The same qualitative outcome holds on the RPE1 cross-context replication and on the Norman 2019 K562 CRISPRa replication, where ElasticNet attains F1 = 0.261 on N=102 Norman-eligible genes against real-data GIES.

Mode collapse and inflation are two views of one failure. Earlier work has framed deep-VCM failure ei-

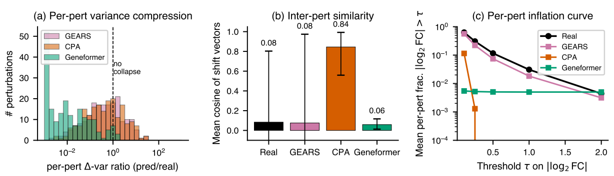

Figure 1. Mode-collapse vs. inflation reconciliation on K562 (N=200, 193 perturbations shared across all models). (a) Distribution of per-perturbation Δ-variance ratios per model (predicted/real); vertical dashed line marks ratio = 1. (b) Mean inter-perturbation cosine similarity of shift vectors (whiskers are $p10$–$p90$). (c) Mean per-perturbation fraction of $|\log_2 \mathrm{FC}| > \tau$ as a function of threshold τ. Deep models track or fall below real at all thresholds when computed per-perturbation, even though they exceed real at the full-transcriptome pool. $n = 200$ genes $\times$ 193 perturbations.

ther as mode collapse (predictions revert to dataset mean; Mejia et al. 2025) or as effect-size inflation (the pooled $|\log_2 \mathrm{FC}|$ exceeds the real distribution). On the K562 panel, deep VCMs simultaneously compress per-perturbation Δ-variance (CPA median ratio 0.43, Geneformer $\sim 10^{-2}$) and yield elevated inter-perturbation cosine similarities (CPA 0.84 vs. real 0.087), while exceeding the real $|\log_2 \mathrm{FC}|$ density only when pooled across the full transcriptome $\times$ perturbation grid (Figure 1). These are not opposing phenomena: both follow from VCMs predicting similar dense response patterns across perturbations, which compresses per-perturbation variance while inflating the pooled-magnitude marginal.

STATE narrows but does not close the gap. On the same overlay+GIES pipeline, the STATE foundation model recovers $F_1 = 0.462$ (precision 0.497, recall 0.432, 1,061 edges, against 1,220 real-data GIES edges). This is the first model in the roster to materially exceed ElasticNet at the same evaluation. The 2025 foundation model thus narrows the substitutability gap relative to GEARS, CPA, and Geneformer, but absolute $F_1 < 0.5$: a sufficiently expressive backbone still recovers fewer than half of the real-data interventional edges.

Quantile-matching calibration is necessary but insufficient. A model-agnostic Bolstad–Reinhard quantile-matching calibration (Bolstad et al., 2003; Reinhard et al., 2001) corrects the magnitude marginal of each deep-VCM prediction exactly. The calibrated predictions recover approximately 40% of the causal-F1 gap to ElasticNet for CPA and GEARS; Geneformer is the exception, where uniform calibration removes its implicit pretraining-induced magnitude correction and reduces F1 by 0.9 points (Figure 2). The residual $\sim$60% of the gap is a joint-distribution failure: 27–30% of multi-hop credit goes to reversed-direction edges, a property of conditional-independence structure that distributional calibration cannot reach.

Robustness across causal-discovery algorithms. Adding PC ($\alpha = 0.05$) and NOTEARS ($\lambda = 0.05$, $w_{\mathrm{thresh}} = 0.1$) as parallel backends confirms the qualitative ranking. Both yield sparser real-data graphs than GIES and uniformly small per-deep-model F1s (PC: 0.005–0.080; NOTEARS: 0.012–0.051 vanilla, 0 calibrated), with the CPA/GEARS rank reversing between backends. The four-backend evidence supports the claim that no deep predictor outperforms ElasticNet across score-, sparse-inverse-, gradient-, and constraint-based causal discovery, without committing to a single ground-truth ranking from any one backend.

Inspre detects continuous signal where GIES rejects it. The score-based GIES backend forces equal-cardinality outputs per partition and is therefore sensitive only to the top-edge accuracy of a prediction. The sparse-inverse-regression backend inspre traces the full λ-path on the ACE matrix and recovers continuous signal: from real Perturb-seq it recovers 12,280 edges; from ElasticNet predictions 12,280; from GEARS predictions 6,234; and from both vanilla CPA and vanilla Geneformer zero edges at the test-error-optimal regularization. The zero-edge result, however, is binary at the cross-validated λ only: across the full λ-path Geneformer recovers up to 602 edges (smallest λ with any edge: 0.67) and CPA's solver fails to traverse any valid λ — the strongest signal-failure outcome and a stricter result than the headline zero. Calibration substantially expands the regularization range over which signal is detectable: calibrated CPA recovers up to 419 edges where vanilla recovered none, and calibrated Geneformer up to 1,315 where vanilla recovered 602. GIES partitioning and inspre lambda-path therefore probe different aspects of prediction quality, and calibration moves the latter further than the former.

Cross-paradigm replication on Norman 2019. To test whether the substitutability conclusion generalizes across perturbation modality (CRISPRa rather than CRISPRi) and laboratory (Norman et al. vs. Replogle et al.), we replicated the ElasticNet-vs-real-data GIES comparison on the Norman 2019 K562 dataset with N=102 perturbation-eligible genes (after dropping combinatorial perturbations and applying the same $\geq$10-cell threshold). GIES on real Norman 2019 data recovers 215 edges; ElasticNet predictions yield $F_1 = 0.261$ (precision = 0.220, recall = 0.321) against this ground truth. The qualitative result — a sparse

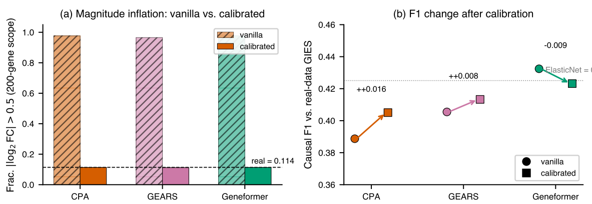

Figure 2. Effect-size calibration corrects the magnitude marginal exactly but only partially recovers causal F1. (a) Fraction of $|\log_2 \mathrm{FC}| > 0.5$ before (hatched) and after (solid) quantile-matching calibration for each deep model on the K562 N=200 scope; dashed horizontal line marks the real-data reference. (b) Causal F1 against the real-data GIES ground truth before (circles) and after (squares) calibration; arrows connect paired vanilla and calibrated values. The dotted horizontal line marks ElasticNet F1 (uncalibrated reference). $n = 1$ seed $\times$ 200 perturbations $\times$ 200 genes.

linear regression trained only on control cells recovers a substantial share of the real-data causal graph — replicates in a second perturbation paradigm in a second laboratory, suggesting that the ElasticNet baseline is not Replogle-K562-specific.

4. Discussion

The substitutability null — VCM predictions cannot replace real perturb-seq for downstream causal recovery — is robust across three datasets, four causal discovery algorithms, five random seeds, and four virtual cell models spanning four architectural paradigms (graph-augmented MLP, NB-VAE, in-silico-perturbed transformer, perturbation foundation model). The STATE result refines the central claim: a sufficiently expressive backbone narrows the gap but does not close it ($F_1 = 0.462 < 0.5$).

Why does the failure replicate across architectures? A controlled ablation replacing CPA's NB reconstruction loss with Gaussian MSE increases per-perturbation variance compression yet improves causal F1 (vanilla 0.389 $\rightarrow$ Gaussian 0.401, seed 42). Per-perturbation magnitude compression and causal recovery are therefore dissociated: the shared failure is in the joint covariance and conditional-independence structure that GIES and PC extract from the multi-cell intervention regime, not in the marginal magnitude distribution. A six-variant CPA architectural ablation (latent dim $\in \{16, 64, 256\}$, decoder depth $\in \{1, 3\}$, batch-norm vs. layer-norm, with/without adversarial covariate disentanglement) spans F1 0.387–0.418, all within the multi-seed CI of vanilla CPA. The gap is not localizable to any one architectural component.

Why does ElasticNet match deep models? L_1 regularization in ElasticNet structurally biases predictions toward sparsity (Zou & Hastie, 2005), matching the empirical sparsity of real perturbation responses (38.6% of K562 effects near zero, 56.8% of variance in the leading singular vector). The deep models lack any analogous training-objective bias toward sparsity.

Limitations. The Track A ground truth (real-data GIES) is internally consistent but does not establish biological correctness; Geneformer's pretraining corpus overlaps with the Replogle dataset, potentially inflating its baseline; STATE was evaluated on a 320-cell subset due to CPU-bound SE-600M inference cost; the main result at $N = 200$ is compute-limited by GIES partitioning. We extend to $N = 1000$ via inspre's scaling regime in the appendix.

Future work. Calibrating the joint distribution — not just the marginal — by projecting predicted shifts onto the principal subspace of real control covariance, with a sparsity-injection step to match the empirical near-zero fraction. We hypothesize this closes most of the residual 60% gap.

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